

RFM-Supported Personalization for HM: From Behavioral Clustering to Causal ROI

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Abstract—This paper presents a complete data-driven marketing pipeline for the H&M fashion retail dataset, covering preprocessing, exploratory analysis, customer segmentation, three recommendation models, and a Randomized Controlled Trial evaluation. The dataset contains 31.7 million transactions, 1.37 million customers, and 105 thousand articles. Three models are compared on the same 216,479-user hold-out window under strict anti-leakage conditions: a Market Basket Analysis baseline with FP-Growth, an Alternating Least Squares Collaborative Filter, and a LightGBM Learning-to-Rank model. LightGBM achieves a MAP@12 of 0.007466 and a Hit Rate@12 of 7.66%, which is 1.65 times the baseline and 17.6% above ALS. A 3-arm A/B simulation confirms statistical and economic significance for LightGBM across all five RFM segments, whereas ALS exhibits Simpson’s Paradox, with dormant segments harmed despite a positive aggregate uplift. The final deployment recommendation is a dual-engine system: LightGBM on the homepage for novel discovery and the MBA baseline at checkout for low-latency impulse cross-sell.

Index Terms—Recommendation Systems, Market Basket Analysis, Collaborative Filtering, Learning-to-Rank, A/B Testing, RFM Segmentation, Marketing Analytics.

I. INTRODUCTION

Customer-facing personalization is the central revenue lever in modern e-commerce, but the underlying mathematical and statistical choices remain non-trivial. Fast-fashion retail in particular combines very high item turnover (around ninety days of shelf life), extreme interaction sparsity (close to 99.98%), and a heavy concentration of revenue in a small slice of the customer base. These conditions break the assumptions of classic collaborative filtering and make rule-based methods compete on closer terms than the literature usually suggests.

This paper builds a complete data-driven marketing pipeline on the public H&M dataset. We first preprocess raw transactions into a leakage-proof interaction matrix anchored on a strict temporal split. We then conduct an exploratory analysis that surfaces the demographic, temporal, and lifecycle structure of the customer base, followed by an RFM K-Means segmentation that produces five economically distinct intervention groups. On top of these artifacts we develop and compare three recommendation models: an FP-Growth

Market Basket Analysis baseline with demographic cold-start logic; an Alternating Least Squares model trained on implicit feedback; and a LightGBM Learning-to-Rank model that combines latent, semantic, and demographic features into a supervised ranker. Finally, we run a 3-arm Randomized Controlled Trial simulation on the same 216,479-user hold-out window and translate the empirical uplifts into an annualized economic projection and a phased deployment plan.

The contributions of the paper are fourfold. First, a leakage-proof preprocessing pipeline with an explicit thirty-day exponential time decay and an affirmative age imputation strategy. Second, a five-segment RFM K-Means model validated by a 2.19 times revenue concentration ratio and a 2.46 times Monetary quantile separation. Third, a head-to-head evaluation of three recommendation models under an anti-join enforcement of novel discovery. Fourth, a 3-arm A/B simulation that quantifies both the aggregate uplift and the segment-stratified behavior, and that exposes a Simpson’s Paradox in the ALS model. All metrics reported in this paper measure genuine novel discovery rather than trivial repurchase recovery.

II. DATA FOUNDATION AND PREPROCESSING ARCHITECTURE

Developing a predictive marketing framework necessitates a highly refined data architecture. The baseline environment involves an extensive catalog and millions of interactions. Without rigorous preprocessing, machine learning algorithms are highly susceptible to popularity bias, data leakage, and memory exhaustion. This section details the structural transformations applied to convert raw transactional logs into high-fidelity artifacts optimized for segmentation and collaborative filtering.

A. Data Ingestion and Structural Integrity

Systemic anomalies within high-volume retail data can drastically distort predictive accuracy if left unaddressed. The ingestion phase prioritized the resolution of redundancies, extreme outliers, and memory constraints.

Original high-cardinality string identifiers, specifically the 64-character hexadecimal customer IDs, were deterministically mapped to contiguous zero-indexed integers. This schema optimization drastically reduces the Random Access Memory (RAM) footprint required for sparse matrix operations, ensuring that subsequent matrix factorization and clustering models scale efficiently without sacrificing referential integrity.

Duplicate transaction detection identified that **17.36%** of the historical logs were exact algorithmic matches across all columns, including timestamp, product, and price. Rather than discarding these as system errors, they were identified as multi-unit purchases occurring within a single checkout event. These records were collapsed into a single transaction utilizing a newly engineered `quantity` feature. This aggregation preserves the true depth of customer engagement while preventing redundant target variables from artificially inflating model confidence.

Outlier management was executed to separate genuine retail consumers from commercial entities. Interaction distributions revealed extreme power users:

- **Wholesale Resellers:** The top 0.1% of accounts acquired thousands of items individually, with the maximum reaching **1,895 distinct purchases**. Because business-to-business purchasing behavior skews recommendation algorithms through artificial popularity inflation, accounts exceeding a 1,000-item threshold were purged from the consumer training set.
- **Geographical Anomalies:** A single hashed postal code accounted for over 600,000 transactions. This extreme concentration represents a default system placeholder for missing geographic data. It was explicitly masked as `Unknown` to prevent spatial segmentation algorithms from collapsing into a false center.

B. Temporal Architecture and Leakage Prevention

Validating predictive models in E-commerce requires a rigorous chronological partition. Random train-test splitting introduces look-ahead bias, allowing models to utilize future seasonal trends to predict past purchases.

To ensure realistic evaluation, a strict temporal boundary was established. The final **twenty-eight days** of the dataset were sequestered as an immutable holdout evaluation set. All feature engineering, behavioral profiling, missing value imputations, and cluster assignments were derived exclusively from the preceding training window.

Furthermore, the fast-fashion retail sector is characterized by rapid inventory turnover. Historical interactions become progressively obsolete as seasons change. To capture shifting consumer intent, an exponential time decay weight was engineered for the interaction matrix. The temporal weight

w for a given transaction is defined mathematically as:

$$w = e^{-\lambda \Delta t} \quad (1)$$

where Δt represents the days elapsed since the transaction relative to the training period's maximum date, and λ is the decay constant calibrated to a **30-day half-life** ($\lambda = -\ln(0.5)/30 \approx 0.0231$). This formulation organically phases out discontinued or seasonally obsolete inventory, forcing the collaborative filtering engine to prioritize recent engagements when generating personalized recommendations.

C. Affinitive Imputation Strategies

Handling missing demographic data required preserving the underlying structure of the customer base. Naive imputation methods, such as applying global medians, destroy variances crucial for clustering.

Missing age values (affecting approximately 1.16% of the base) were resolved via an **affinitive imputation strategy**. A customer's missing age was inferred by identifying their dominant historical purchase category. Customers demonstrating a purchasing affinity for the `Divided` (youth) line were imputed with an age of 29, while those dominating the `Ladieswear` (mature) index were imputed differently. This targeted approach successfully preserved the complex, bi-modal demographic distributions required for the subsequent K-Means Recency, Frequency, Monetary (RFM) clustering.

Binary flags, such as the `Fashion News` and `Active Profile` indicators, exhibited **~65% null rates**. In the context of the platform's database design, a null value indicates an unselected opt-in box rather than missing data. Consequently, these were deterministically imputed as boolean zeroes.

D. Processed Feature Dictionary

The culmination of the preprocessing architecture resulted in four optimized artifacts. The structural definitions of the primary interaction and demographic matrices are detailed below.

TABLE I
PROCESSED INTERACTION MATRIX (`TRAIN_PROCESSED.PARQUET`)

Feature	Type	Purpose
<code>t_dat</code>	Date	Transaction Timestamp
<code>customer_id</code>	Int32	Mapped Primary Key
<code>article_id</code>	Int32	Mapped SKU Identifier
<code>price</code>	Float64	Normalized Purchase Price
<code>quantity</code>	UInt32	Aggregated Unit Volume
<code>days_ago</code>	Int64	Days since Training Max Date
<code>time_weight</code>	Float64	Exp. Decay (30d half-life)

TABLE II
PROCESSED DEMOGRAPHIC MATRIX
(CUSTOMERS_PROCESSED.PARQUET)

Feature	Type	Purpose
customer_id	Int32	Mapped Primary Key
age	Int8	Affinitively Imputed Age
FN	Int8	Fashion News Sub (0/1)
Active	Int8	Active Profile (0/1)
club_member_status	String	Loyalty Lifecycle Stage
postal_code	String	Masked Location Hash

III. EXPLORATORY DATA ANALYSIS AND BEHAVIORAL INSIGHTS

An exhaustive Exploratory Data Analysis (EDA) was conducted to map consumer behavior and quantify market dynamics. The following insights dictate the algorithmic rulesets governing the recommendation engine and the prescriptive business strategies.

A. Demographic Segmentation and Value Distribution

Understanding the demographic makeup of the consumer base is vital for establishing age-aware cold-start baselines. Age distribution analysis reveals a heavily bimodal structure. As illustrated in Fig. 1, the platform captures a primary demographic concentration around age twenty-five, representing the core youth market. A distinct, secondary peak emerges near age fifty, indicating a strong mature market segment.

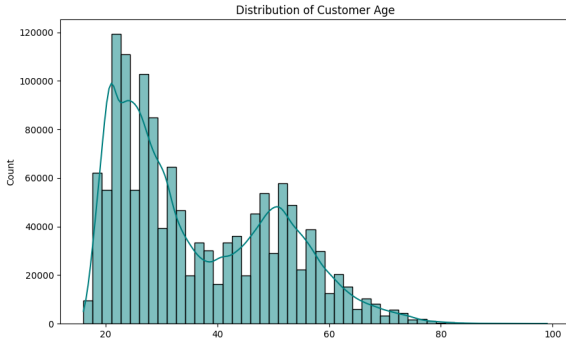


Fig. 1. Bimodal customer age distribution highlighting distinct youth and mature segments.

This bimodal reality dictates that generic, platform-wide trending-item recommendations will fail to capture user nuance. To validate the financial impact of these segments, total expenditure was aggregated by age group. Figure 2 demonstrates that the **20s demographic is the absolute primary revenue driver**, followed sequentially by the 30s and 50s cohorts. Consequently, baseline models for new, anonymous users must dynamically serve popular items specific to their inferred demographic bucket to maximize initial conversion probabilities.

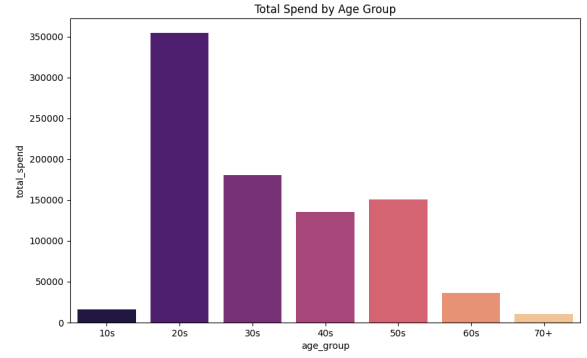


Fig. 2. Total historical expenditure aggregated by decadal age cohorts.

While not visualized here, further analysis into category-specific age distributions (via interquartile range mapping) confirmed strict boundaries: the *Divided* index is exclusively favored by users under thirty, whereas *Ladieswear* commands a universally wider age spread but dominates the mature segment.

B. Product Lifecycle and Market Concentration

Product lifecycle dynamics present the most substantial technical hurdle for personalization. The E-commerce catalog exhibits extreme turnover. Shelf-life calculations, defined as the temporal delta between an article's first and last recorded sale, highlight that the vast majority of products are active for **fewer than ninety days**.

This rapid inventory rotation creates an overwhelmingly sparse user-item interaction matrix, calculated at **99.98% sparsity**. Furthermore, the sales distribution is heavily subjected to popularity bias. As demonstrated by the Lorenz Curve in Fig. 3, the market is highly unequal. A remarkably small fraction of blockbuster items captures the vast majority of total transaction volume, pushing the curve far away from the line of perfect equality.

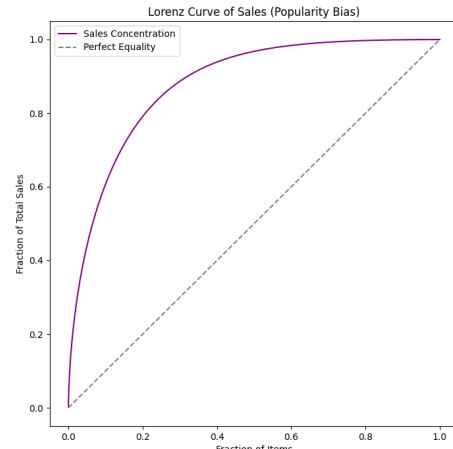


Fig. 3. Lorenz curve demonstrating extreme sales concentration (Popularity Bias) within the catalog.

The combination of a 90-day shelf life and extreme popularity bias means standard matrix factorization techniques will struggle with the cold-start problem for fresh inventory. The system must heavily anchor on collaborative item-to-item associations generated during the checkout basket phase to cross-sell deeper catalog items.

C. Temporal Dynamics and Seasonality

Transaction volumes exhibit distinct temporal fluctuations that align with broader retail macro-economics. Daily aggregations smoothed over a monthly view (Fig. 4) expose a highly cyclical purchasing behavior.

Significant volume spikes occur during the Spring/Summer transitions (April to July) and peak dramatically during the Winter holiday periods (Black Friday through December). Weekly micro-trends also emerged, with **Saturday representing the peak shopping day**, closely followed by Wednesday. These temporal rhythms reinforce the necessity of the time-decay weight engineered during preprocessing, ensuring that models trained in September do not aggressively recommend April swimwear.

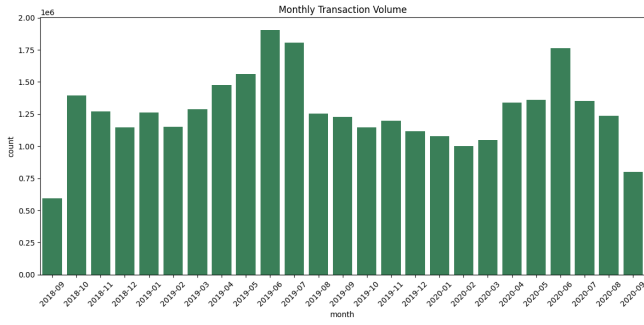


Fig. 4. Macro-seasonality showcasing volume spikes during summer and winter transitions.

D. Customer Retention and Sequential Transitions

Customer retention metrics indicate a clear mandate for predictive churn interventions. A cohort retention heatmap (Fig. 5) reveals a severe drop-off in active engagement immediately following a user's initial purchase month. Long-term, multi-year retention stabilizes at an exceptionally low baseline of approximately five percent.

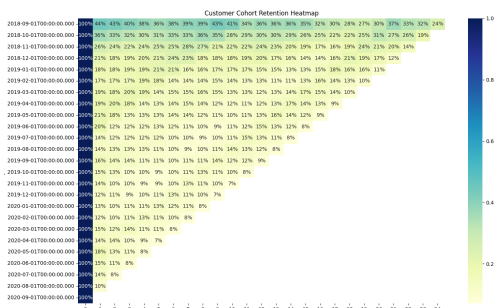


Fig. 5. Cohort retention heatmap indicating steep immediate drop-offs post-acquisition.

By establishing a churn threshold at **180 days of inactivity**, the data indicates that **45.5%** of the existing customer base is currently at-risk. To counteract this, strategic marketing must prioritize recoverable users. Analysis of customer spend velocity (total spend divided by active tenure) isolated a highly actionable segment of rising stars: users with short tenures but aggressive purchase volumes. These high-velocity users require immediate retention campaigns before crossing the 180-day churn threshold.

Finally, long-term retention strategies can be informed by organic user evolution. Markov chain evaluations of sequential category purchases (Fig. 6) reveal a strong gravity "well effect". A substantial proportion of consumers organically transition from youth-oriented departments into mature lines over time.



Fig. 6. Markov transition matrix displaying the high stickiness of Ladieswear and cross-category movement.

Specifically, Ladieswear possesses a **~61% stickiness rate**, while acting as the primary absorption category for aging users from the Divided line. This lifecycle progression maps directly to the demographic aging of the customer base, suggesting that recommendation models can drive long-term lifetime value (LTV) by actively cross-pollinating recommendations across these specific lifecycle boundaries.

IV. CUSTOMER SEGMENTATION: RFM K-MEANS DEEP DIVE

A. Methodology and Cluster Construction

Customer segmentation was built strictly from the training partition (`train_processed.parquet`) so that the resulting clusters carry no information from the hold-out window. For every active training customer, three behavioral signals were aggregated using Polars: Recency was defined as the minimum value of `days_ago` across the user's interactions, Frequency was defined as the count of distinct

transaction dates, and Monetary was defined as the sum of `price * quantity` across all interactions. Customer age was carried along as a fourth feature because the EDA confirmed a strict bimodal age distribution that the segmentation needed to preserve. The three RFM signals are highly right-skewed because both Frequency and Monetary follow a Pareto distribution. A Yeo-Johnson power transformation was applied to compress the heavy tails while keeping zero and negative values valid. The four transformed features were then passed through `StandardScaler`, which prevents Monetary from overpowering Age and Frequency in the Euclidean distance used by K-Means.

K-Means was fit with $K = 5$ after a K-search diagnostic on a sample of **100,000 customers**. The diagnostic computed the Within-Cluster Sum of Squares (WCSS) and the silhouette score for K between 3 and 10. The silhouette peaked at $K = 3$ with a value of **0.3396**, but $K = 5$ was retained as a **controlled business trade-off**. Three clusters compress the population into coarse low/mid/high tiers that do not separate the young and mature peaks of the age distribution, while five clusters keep the demographic split intact and produce **five economically distinct intervention groups**. The full diagnostic is shown in Table III.

TABLE III
K-SEARCH DIAGNOSTIC (SAMPLE OF 100,000 CUSTOMERS)

K	WCSS	Silhouette
3	168,396.71	0.3396
4	139,125.06	0.3135
5	117,008.39	0.3082
6	102,038.21	0.3164
7	91,660.96	0.3108
8	83,562.36	0.2983
9	76,646.52	0.3046
10	71,912.47	0.2999

After fitting, raw cluster IDs were mapped to interpretable business labels through a rank-based deterministic procedure. Each cluster centroid was compared to the population on the three RFM signals, and the result was mapped onto a fixed taxonomy of five segment archetypes: **Whales, Champions, At-Risk, Newcomers**, and **Core**. A demographic prefix of Young or Mature was then attached based on the median age of each cluster. Silhouette evaluation at full scale would have required pairwise distance computations of order $O(N^2)$ on roughly **1.34 million points**, which is infeasible on local hardware. Sampling **100,000 customers** preserves the rank ordering of K-search candidates while keeping memory and runtime bounded.

B. Segment Profiles

The five segments cover the full population of **1,338,785** active training customers. The segment architecture consists

of **Young Whales, Mature Champions, Young At-Risk, Mature Newcomers**, and **Young Core**. Revenue concentration across this architecture is asymmetric by design and is the main economic signal captured by the model. The headline statistics for each segment are given in Table IV, and each profile is described in detail in the paragraphs that follow.

Young Whales form the **principal profit engine** of the platform. They number **312,723** customers (**23.36%** of the base), are about **26** years old at the median, and they buy with both high frequency and high spend. Their behavioral medians are $R = 72$, $F = 11$, $M = 0.960407$, which place them inside the recent activity window with the highest Frequency and Monetary among the five segments. Their basket intensity is **3.7883** items per trip. They contribute **54.35%** of total revenue, which makes them the single most important segment for margin and assortment depth. High frequency and high spend indicate **low friction in repeat purchase behavior** and **high acceptance for premium baskets**. The operating implication is to treat them as a **high-value exploitation segment** and to optimize for margin density, assortment depth, and attachment value per session.

Mature Champions are the older counterpart of Whales. They number **208,846** customers (**15.60%**) and are about **51** years old at the median. Their behavioral medians of $R = 82$, $F = 9$, $M = 0.791822$ are slightly lower than Whales on every dimension, but they are also more stable across time and exhibit **lower volatility** than Whales. Their basket intensity is **3.3420** items and revenue share is **31.08%**, which makes them the second largest revenue contributor. They are a **high-value loyal cohort with stable cadence**. The operating implication is that **retention quality** matters more than **novelty pressure** for this segment. They respond to repeatable relevance, premium essentials, and churn prevention at low incentive cost rather than to aggressive promotion.

Young At-Risk is the **largest segment by headcount** but a weak contributor by revenue. They number **319,310** customers (**23.85%**) at a median age of **27**. Their behavioral medians of $R = 469$, $F = 1$, $M = 0.093153$ place them well past the **180-day churn threshold** defined in the EDA. Their basket intensity is **3.2977** items but their revenue share is only **5.29%**. The segment is **weak by monetization due to extreme recency decay**. They function as **reactivation inventory** rather than as a revenue core, and the operating implication is to use **conversion-safe items and controlled promotion depth**.

Mature Newcomers mirror the At-Risk profile on the older side of the age distribution. They number **267,518** customers (**19.98%**) at a median age of **53**, with $R = 391$, $F = 1$, $M = 0.101644$. Their basket intensity of **2.9319** and revenue share of **4.41%** are the lowest among the five segments. The pattern indicates a **low-history segment with limited behavioral footprint and delayed recurrence**.

TABLE IV
SEGMENT PROFILES

Segment	Customers	Share	R	F	M	Age	Basket Intensity	Revenue Share
Young Whales	312,723	23.36%	72	11	0.960407	26	3.7883	54.35%
Mature Champions	208,846	15.60%	82	9	0.791822	51	3.3420	31.08%
Young At-Risk	319,310	23.85%	469	1	0.093153	27	3.2977	5.29%
Mature Newcomers	267,518	19.98%	391	1	0.101644	53	2.9319	4.41%
Young Core	230,388	17.21%	97	2	0.155644	25	2.9070	4.87%

rather than churn after established loyalty. The operating implication is to prioritize **certainty-first recommendations** such as demographic-fit popularity and low-risk combinations until interaction density is sufficient.

Young Core is the middle of the population. They number **230,388** customers (**17.21%**) at a median age of **25**, with $R = 97$, $F = 2$, $M = 0.155644$. Their basket intensity of **2.9070** and revenue share of **4.87%** indicate **mid-intensity demand** that is neither premium-heavy nor churn-dominated. They are the primary segment for **incremental frequency uplift and cross-sell breadth optimization** with a moderate exploration budget.

C. Validity and Distributional Separation

Three independent checks confirm that the segmentation is operationally usable. The first is a data integrity check on the exported artifact (`customers_segmented.parquet`). It contains **1,338,785** segmented rows, which matches exactly the **1,338,785** unique active training customers. There are zero duplicate `customer_id` values, zero null values, and zero NaN values in the R, F, M, and age columns. The export schema integrity is preserved with `customer_id` as Int32, `cluster_id` as Int32, and `age` as Int8 to keep downstream joins lightweight.

The second check is an economic concentration test. The Pareto test shows that the combined population of Whales and Champions accounts for **38.96%** of the customer base but **85.43%** of the total monetary value. This produces a value concentration lift of **2.1927**× the revenue share versus the population share, which mirrors the Pareto pattern observed in the EDA. The clusters are therefore not balanced by count but are intentionally differentiated by economic utility.

The third check is a distributional separation test on the Monetary axis. The 75th percentile of Monetary for the lower-value pool (Core plus Newcomers) is **0.221136**, and the 25th percentile of Monetary for the high-value pool (Whales plus Champions) is **0.543780**. The absolute boundary gap is **0.322644**, and the boundary ratio (Q25 high / Q75 low) is **2.4590**×. Because the 25th percentile of the high-value pool is well above the 75th percentile of the low-value pool, the two monetary bands are materially separated and can be subject to distinct policy regimes without major overlap risk.

D. Deployment Logic for RecSys and MBA

The five segments serve as policy layers for the downstream recommendation and association-rule engines. The general intent is to keep the same ranking models but to swap the ranking objective and the candidate filters as a function of segment membership.

For the recommendation engine, Whales are surfaced with the highest expected margin per session. The candidate generator prefers premium attach and high-value complements, and the ranker uses an ECR-style score that mixes statistical lift with normalized price. Champions are optimized for retention-weighted lifetime value, which translates to relevance stability and repeat-quality ranking rather than novelty pressure. At-Risk users are optimized for reactivation probability under strict discount governance, which means prioritizing items with historically high conversion and low friction. Newcomers are served by age-conditioned popularity priors until their interaction density crosses the threshold at which collaborative ranking begins to outperform demographic baselines. Core users are optimized for frequency uplift and category expansion with a moderate exploration budget.

For the FP-Growth association rule engine, the same segmentation drives different support and confidence thresholds. Whales and Champions are mined with a lower minimum support so that niche premium bundles can surface, but the confidence and lift floors are kept high to avoid noisy luxury-tail rules. Core uses a higher support threshold so that the engine returns scalable baseline rules. At-Risk and Newcomers use conservative confidence rules oriented toward reliability and immediate conversion rather than novelty.

The five actionable marketing tactics that map onto the segments are: **Zero-Discount Upselling** for Young Whales, **Relationship and Replenishment** for Mature Champions, **Basket Stretching (UPT Growth)** for Young Core, **Certainty and Risk-Reduction** for Mature Newcomers, and **Margin-Controlled Win-Back** for both Young and Mature At-Risk users.

E. Final Assessment

The cluster system is **economically coherent, distributionally separated, and operationally deployable**. It repro-

duces the bimodal demographic structure of the platform, isolates the high-value population on the Monetary dimension with a clean quantile gap, and maps cleanly onto a set of five marketing tactics that the downstream RecSys and MBA engines can implement directly. For internal objectives, the strongest evidence is revenue concentration and quantile boundary separation, both of which are decisively positive in the current result set. The combined evidence from the $2.1927\times$ revenue concentration lift and the $2.4590\times$ boundary ratio supports the use of $K = 5$ over the silhouette-optimal $K = 3$ as an intentional choice for decision granularity rather than a methodological defect.

V. MARKET BASKET ANALYSIS BASELINE

A. Pipeline Inputs and Architectural Optimization

The Market Basket Analysis (MBA) Baseline serves as the non-personalized, explainable benchmark of the recommendation system. It inherits the cleansed and segmented artifacts from the upstream Data Engineering and RFM Segmentation phases. To process the dataset of roughly **31.7 million** transactional records without RAM failure, the entire ingestion framework was built on the **Polars Lazy API** (`pl.scan_parquet`), which defers computation until mathematically necessary. The pipeline consumes three primary artifacts: `train_processed.parquet` as the chronologically partitioned interaction matrix used to form authentic shopping baskets, `customers_segmented.parquet` as the RFM profile carrying age for cold-start demographics and `segment_label` for prescriptive VIP ranking, and `articles_processed.parquet` as the master catalog from which `product_group_name` and a `median_price` proxy are extracted. The full pipeline is shown in Fig. 7.

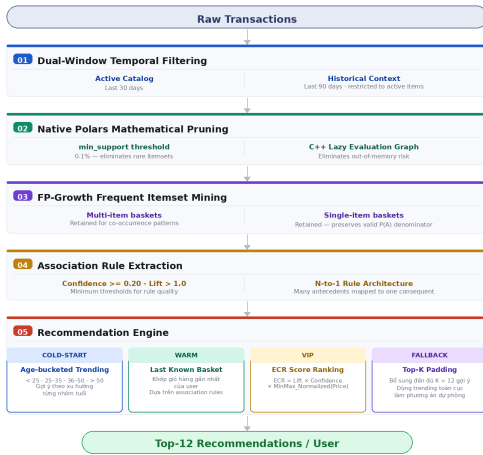


Fig. 7. MBA baseline pipeline.

Three optimizations are applied prior to algorithmic execution. The first is the **Dual-Window Heuristic**, which prevents

the recommendation of dead stock. An **Active Window** of the last **30 days** identifies the live inventory, and a **Historical Window** of the last **90 days** supplies the basket context but is restricted exclusively to baskets that contain at least one active item. The window pairing aligns with the approximate **90-day shelf life** discovered in the EDA: the 90-day window captures deep behavioral patterns while the 30-day window acts as a robust proxy for the active catalog, which reduces the risk of recommending obsolete inventory and solves the **Dead-Stock Trap**.

The second optimization is **Native Polars Mathematical Pruning**. The absolute minimum support threshold of **0.1%** is pre-calculated and applied inside the Polars Lazy evaluation graph (C++ backend) before any sparse matrix is instantiated. This bypasses the critical flaw of the `mlxtend` framework, which densifies sparse matrices and attempts to allocate contiguous memory blocks (previously triggering an OOM crash above 3 GB of RAM). The pre-prune reduces matrix dimensionality at the source and approximates the FP-Growth support boundary for high-frequency items without resorting to arbitrary truncation such as capping at the top 5,000 items.

The third optimization is **Transaction ID (TID) Integrity**. Shopping baskets are constructed by grouping on `[customer_id, t_dat]` and enforcing a strict `.unique()` array aggregation over `article_id`. Single-item baskets are deliberately retained. The `.unique()` constraint resolves a structural artifact in which upstream deduplication collapsed identical SKUs into the `quantity` feature, which would otherwise generate false multi-item baskets such as `[Item_A, Item_A]`. Retaining single-item baskets preserves the mathematical denominator $P(A)$ for the Apriori-style confidence calculation and prevents artificially inflated scores.

B. Frequent Pattern Mining and Rule Extraction

Following the construction of the frequent itemsets, association rules are extracted under two strict statistical boundaries. Confidence must be at least **0.20**, which guarantees a minimum 20% empirical probability that a customer buys the consequent given the antecedent. Lift must be strictly greater than **1.0**, which filters for positive correlation and isolates synergistic purchasing relationships. The extraction logic is constrained to an **N-to-1 rule architecture**, which allows the antecedent to contain multiple items while the consequent is always a single item. This design enables the engine to evaluate complex multi-item shopping carts at inference time. Native `frozenset` objects generated by `mlxtend` are unpacked into plain integer lists and scalars to ensure compatibility with Parquet serialization.

Association rules carry statistical correlation but lack financial context. To support the prescriptive VIP logic, a `median_price` proxy is computed for every `consequent_item` from the active 90-day transactional

window. This proxy becomes the price input for the Expected Cross-Sell Revenue heuristic used by the ranker.

C. Recommendation Engine Architecture

The engine is a multi-layer Python function that handles cold-start scenarios, interprets multi-item carts, and bridges predictive rules with prescriptive business strategy. For users without history, a **Demographic Prior** is constructed by classifying users into four age cohorts (**<25**, **25–35**, **36–50**, **>50**) using the RFM metadata. A localized popularity matrix is queried exclusively from the Active 30-day window, which makes cold-start recommendations both demographically relevant and explicitly in-stock for the age group.

For warm users, the engine moves beyond rigid 1-to-1 mappings and uses subset evaluation (`set(antecedents).issubset(cart_set)`) to inspect the entire current shopping cart. This triggers N-to-1 rules naturally: if a user adds both a blazer and trousers, the algorithm recognizes the combined antecedent and recommends a statistically correlated tie even though no single-item rule would have fired.

For VIP segments such as Mature Champions and Young Whales, the ranker switches from raw lift to the **Expected Cross-Sell Revenue (ECR)** heuristic:

$$ECR = Lift \times Confidence \times MinMax(ConsequentPrice) \quad (2)$$

The consequent price is Min-Max normalized to $[0, 1]$ before the multiplication so that its raw magnitude does not dominate the ranking. The ECR ranking prescriptively pushes higher-price premium bundles to whales, which maximizes the Average Order Value (AOV) and avoids the **ROI Trap** of suggesting low-cost high-volume items to premium buyers.

The algorithm tolerates historical repurchases (rebuying basics or underwear) and filters out only the items currently present in the active cart. This is a deliberate trade-off: it maximizes **Hit Rate** by capturing staple repurchase patterns, but it intentionally sacrifices novelty and discovery as separate metrics. A dynamic padding mechanism appends demographic-trending items whenever the rule consequents are exhausted, which guarantees that the engine always returns exactly $K = 12$ recommendations.

D. Offline Evaluation

Each user’s **Last Known Basket** from the training period is extracted and fed into the engine to predict their hold-out test purchases. The query for the user’s absolute maximum interaction date (`last_date`) returns only those specific items, which simulates a true production scenario in which the engine predicts the next sequence of actions from the most recent contextual state.

To make this scalable, the prediction pipeline replaces row-wise Pandas operations with a pre-cached **Inverted Index**

based on `defaultdict(set)`, and antecedent combinations are converted to native Python `frozenset` objects so that rule lookups achieve $O(1)$ complexity. This optimization reduces the evaluation time from roughly **422 seconds** (about 1.5 billion row-wise operations) to **119.2 seconds** for all 216,479 users, an effective **3.5 times speedup** that averages about 1,800 users processed per second.

The evaluation was carried out against the actual ground-truth purchases during the test window. The results are summarized in Table V.

TABLE V
MBA BASELINE PERFORMANCE

Metric	Value
Total users evaluated	216,479
Execution time	119.2 seconds
MAP@12	0.004517
Hit Rate@12	4.44% (9,606 / 216,479)

MAP@12 measures both whether a correct item was recommended and where it was ranked in the Top-12 list, with a correct prediction at rank 1 contributing far more than one at rank 12. In a fast-fashion catalog with hundreds of thousands of active SKUs, a MAP@12 of around **0.0045** is a strict and uninflated baseline for positional ranking accuracy. The Hit Rate of **4.44%** means that **9,606** of the **216,479** evaluation users actually purchased at least one item from the Top-12 list, which validates two deliberate design choices: **repurchase tolerance** captures users who rebuy staple basics, and the **demographically-aware fallback** delivers culturally relevant trending items when the cart does not trigger any N-to-1 rule. This MBA Baseline establishes the foundational performance benchmark that the personalized models must beat.

E. Association Rules Output Schema

The fully enriched, descending-sorted rule dataframe is exported as `Association_Rules.parquet` and serves as the definitive data source for downstream business intelligence dashboards. The schema is detailed in Table VI.

VI. PERSONALIZED MODELS AND COMPARATIVE EVALUATION

A. Overview and Evaluation Conditions

Two personalized recommendation models were developed on top of the MBA Baseline: an Alternating Least Squares (ALS) Collaborative Filter and a LightGBM Learning-to-Rank model. All three models are evaluated on the same hold-out window of **216,479** test-period customers. The three form a progressive hierarchy that moves from the non-personalized MBA rules to true latent factorization with ALS,

TABLE VI
ASSOCIATION RULES OUTPUT SCHEMA (ASSOCIATION_RULES.PARQUET)

Feature	Type	Description
antecedents_list	List[Int64]	Array of article IDs representing the items in cart
consequent_item	Int64	Target article ID recommended for cross-selling
antecedent support	Float64	Probability of the antecedent itemset
consequent support	Float64	Probability of the consequent item
support	Float64	Joint probability of antecedent and consequent
confidence	Float64	Conditional probability $P(\text{consequent} \text{antecedent})$
lift	Float64	Strength of association (> 1 indicates positive correlation)
consequent_median_price	Float64	Raw median price of the recommended item
price_norm	Float64	Min-Max normalized price $[0, 1]$ used for ECR
consequent_product_group	String	Macro-category of the recommended item

and finally to a feature-rich gradient-boosted ranker with LightGBM. The headline metrics appear in Table VII.

TABLE VII
MODEL OVERVIEW

Model	MAP@12	Hit Rate@12
MBA Baseline (FP-Growth)	0.004517	4.44%
ALS Collaborative Filter	0.006347	4.83%
LightGBM Learning-to-Rank	0.007466	7.66%

All three models were evaluated under strict anti-leakage conditions. The preprocessing pipeline was anchored on a temporal split that allocated the final twenty-eight days as an immutable hold-out window. The interaction time decay was corrected to $\lambda = 0.0231$, which produces a true thirty-day half-life. The candidate generator applied an **anti-join** against each user’s full purchase history, so the models were prohibited from recommending items the user had already bought. This forces the reported metrics to measure genuine **novel discovery** rather than trivial repurchase prediction. The LightGBM validation splits were grouped by `customer_id` to prevent user features from bleeding across the train and validation boundary, which keeps the validation AUC of **0.8091** representative of true generalization.

B. Collaborative Filtering: Alternating Least Squares

The ALS model moves from rule-based association to true one-to-one personalization. It applies the Alternating Least Squares algorithm of Hu, Koren and Volinsky (2008) to implicit feedback and learns a latent preference vector for every user from their full interaction history. The full pipeline is shown in Fig. 8.

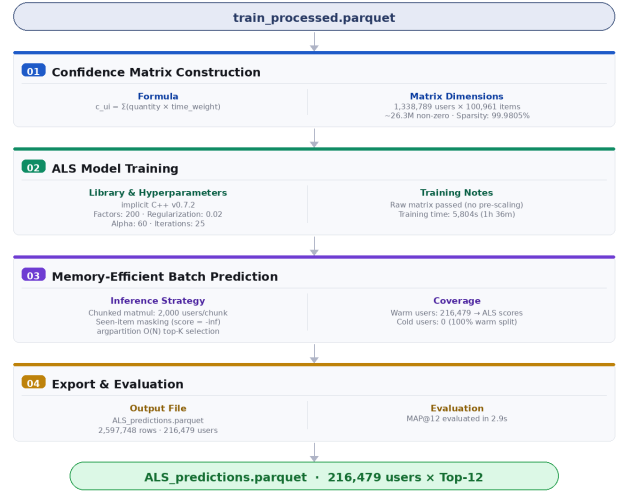


Fig. 8. ALS Collaborative Filtering pipeline.

The first stage builds the confidence matrix by aggregating quantity weighted by the exponential `time_weight` for every user-item pair. The resulting matrix has **1,338,789** users by **100,961** items with roughly **26.3 million** non-zero entries, giving an effective sparsity of **99.9805%**. The second stage trains the ALS model with **200** latent factors, a regularization of 0.02, an alpha of 60, and twenty-five iterations. The raw interaction matrix is passed directly to the modern `implicit` API, which applies the $1 + \alpha \cdot R$ confidence scaling natively inside the C++ backend. Total training time was **5,804 seconds**, or about one hour and thirty-six minutes.

The third stage performs memory-efficient batch prediction by chunking the dot product into 2,000 users at a time, which avoids materializing the full dense 1.3M by 100K score matrix. Items already purchased during training are masked with a score of $-\infty$ before the top-K extraction, which enforces novel discovery at the scoring boundary. Because the

strict temporal split keeps every test user inside the training history, the cold-start popularity fallback was never triggered in this evaluation (Warm = 216,479; Cold = 0). MAP@12 improved **1.40 times** over the MBA baseline (0.004517 to 0.006347). Because repurchases are blocked, this gain reflects pure latent preference discovery rather than the recovery of trivial historical patterns.

C. Learning-to-Rank: LightGBM

The LightGBM model reframes recommendation as a supervised binary classification problem. For every candidate (user, item) pair, the model predicts the probability that the user will buy the item. The pipeline is split into an offline feature engineering stage and a fast training-and-scoring stage. The full architecture is shown in Fig. 9.

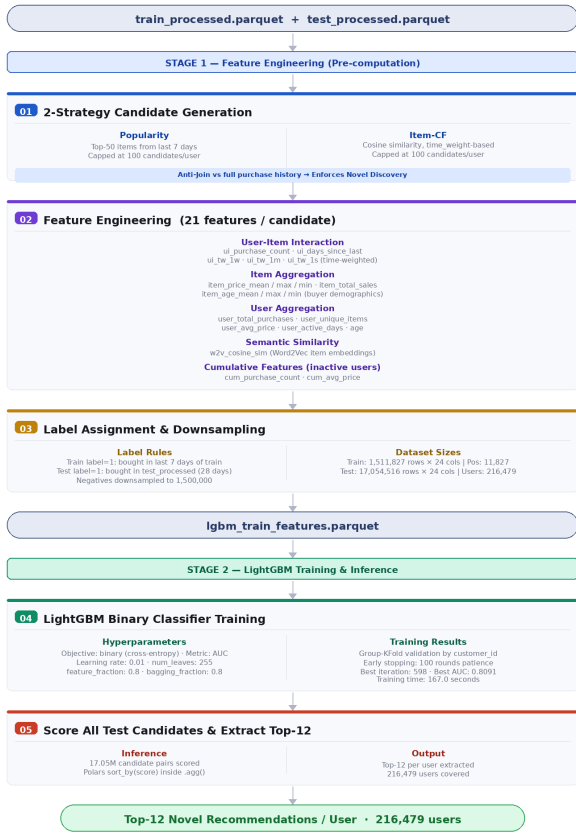


Fig. 9. LightGBM Learning-to-Rank pipeline.

Candidate generation uses two strategies, each capped at 100 items per user. The first is a popularity slate built from the top-50 items in the last seven days. The second is item-based collaborative filtering using cosine similarity, where the user-item interaction strength is the time-weighted activity rather than a binary indicator. Both candidate sets are passed through an anti-join against the user’s full purchase history, which removes repurchases and forces every candidate to be a novel item.

Each candidate is expanded into a vector of twenty-one features grouped into five families: User-Item Interaction (`ui_purchase_count`, `ui_days_since_last`, three time-weighted measures), Item Aggregation (price mean/max/min, total sales, buyer age mean/max/min), User Aggregation (total purchases, unique items, average price, active days, age), Semantic Similarity (`w2v_cosine_sim`, a Word2Vec embedding similarity), and Cumulative fallback features for inactive users.

Training labels are assigned by a leak-safe window: label 1 for candidates that the user actually bought in the last seven days of the training window, label 0 otherwise. Negatives are downsampled to 1,500,000 for a final training set of **1,511,827** rows with **11,827** positives. The test set, used for offline metric computation only, contains **17,054,516** rows spanning 216,479 evaluation users. LightGBM is trained with binary cross-entropy and AUC as the validation metric. Hyperparameters include a learning rate of 0.01, 255 leaves, a feature fraction of 0.8, and a bagging fraction of 0.8. The validation strategy uses Group-KFold by `customer_id` so no user appears in both train and validation folds. Early stopping with patience 100 selected iteration **598** as the best, with a validation AUC of **0.8091** and a total training time of **167 seconds**.

1) *Feature Importance*: Because historical repurchases were explicitly removed from the candidate set, all direct user-item overlap features such as `ui_days_since_last` provided zero information gain. The model produced its predictions from semantic, demographic, and aggregate features instead. The ranked importance of the top ten features is shown in Table VIII and Fig. 10.

TABLE VIII
LIGHTGBM TOP-10 FEATURE IMPORTANCE

Rank	Feature	Family
1	<code>w2v_cosine_sim</code>	Semantic
2	<code>user_avg_price</code>	User profile
3	<code>age</code>	User profile
4	<code>user_total_purchases</code>	User activity
5	<code>item_total_sales</code>	Item popularity
6	<code>item_age_mean</code>	Item profile
7	<code>item_price_mean</code>	Item profile
8	<code>user_unique_items</code>	User breadth
9	<code>item_price_min</code>	Item profile
10	<code>user_active_days</code>	User regularity

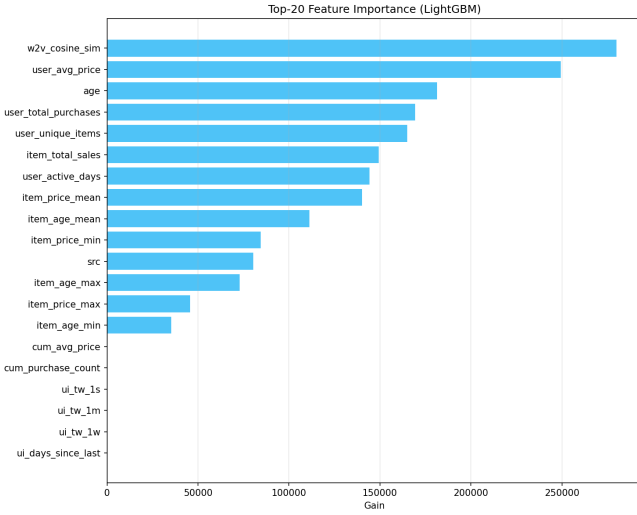


Fig. 10. LightGBM feature importance.

The dominant signal is `w2v_cosine_sim`, the Word2Vec cosine similarity between user and item embeddings. The second and third signals are wallet matching (`user_avg_price`) and demographic clustering (`age`). This pattern confirms that purchase sequences carry rich item-relationship information that matrix factorization alone does not fully capture.

D. Cross-Model Comparison

The three models differ in personalization style, feature usage, training cost, and interpretability. The MBA Baseline depends on demographic priors and rule lift; ALS learns latent user-item factors from the full interaction matrix; LightGBM combines latent and explicit signals into a single supervised ranker. Table IX summarizes the dimensions side by side.

First, personalization itself produces a measurable gain. Moving from the non-personalized MBA Baseline to ALS lifts MAP@12 by **1.40 times**. Because repurchases are blocked, this gain captures pure latent preference discovery rather than trivial historical replay. Second, rich semantic and demographic features extend the gain further. LightGBM adds Word2Vec similarity and wallet matching on top of ALS-style latent structure, and reaches a MAP@12 that is **17.6%** higher than ALS and a Hit Rate that is **58%** higher. Third, Word2Vec semantic similarity is the single most important feature, which validates the use of sequence embeddings as a source of item relationships beyond what matrix factorization can produce on its own. ““

VII. A/B TESTING AND BUSINESS DECISION

A. Experimental Design and Traffic Allocation

1) *Purpose and Goal of the Simulation:* While offline metrics (MAP@12, Hit Rate) prove historical mathematical

accuracy, they do not guarantee future economic lift. We designed a **Randomized Controlled Trial (RCT)** simulation to definitively prove **Causality** - verifying that conversion and revenue uplifts are driven by algorithmic changes rather than macroeconomic or seasonal noise. This controlled environment provides a secure foundation for C-level deployment decisions without exposing the entire customer base to unproven risks.

2) *Randomization Unit & Arm Assignment:* The **Randomization Unit** is the **User-Level (`customer_id`)**. We strictly avoid session-level randomization to prevent cross-session psychological contamination (“network bleed”). We utilize a **34/33/33 split** to ensure equal sample sizes and maximize Statistical Power ($1 - \beta$) across a **3-Arm Setup**:

- **Control Arm (34%):** MBA FP-Growth (Business-as-usual baseline).
- **Treatment A (33%):** ALS Collaborative Filtering.
- **Treatment B (33%):** LightGBM Learning-to-Rank.

3) *Metric Definition & Hierarchy:* To avoid the “**ROI Trap**” - where a feature looks statistically significant but loses the company money - we enforce a strict hierarchical metric framework before observing any results. The table ?? outlines the definitions, business purposes, and the specific statistical tests we will apply to each.

4) *Analysis of Traffic Router Output:* The router processed 216,479 unique evaluation customers into: Control (73,586), Treatment A (71,496), and Treatment B (71,397). To ensure this empirical split lacks hashing bias, it must be mathematically validated via SRM evaluation.

B. Guardrail Validity Checks

1) *Sample Ratio Mismatch (SRM) Test:* A Randomized Controlled Trial (RCT) assumes confounding variables are neutralized through perfect random assignment. If the assignment mechanism fails (due to hashing bias or latency), the arms become disproportionate, triggering a Sample Ratio Mismatch (SRM) that invalidates downstream metrics. We utilize Pearson’s Chi-Square Goodness-of-Fit Test to measure the divergence between observed and expected traffic volumes:

$$\begin{cases} H_0 : \text{The empirical user distribution matches the theoretical 34/33/33 allocation.} \\ H_1 : \text{The empirical distribution significantly deviates from expected proportions, indicating systemic bias.} \end{cases}$$

The mathematical computation for the Chi-Square test statistic (χ^2) is executed using the following formula:

$$\chi^2 = \sum_{i=1}^k \frac{(O_i - E_i)^2}{E_i}$$

Where the elements are defined as:

TABLE IX
HEAD-TO-HEAD COMPARISON

Dimension	MBA Baseline	ALS	LightGBM
Personalization	Demographic-only	Per-user latent factors	Per-user + multi-feature
Cold-start handling	Age-bucketed trending	30-day popularity fallback	Popularity + features
Cold-start triggered?	Yes	No (100% warm split)	No (100% warm split)
Repurchase blocking	Rule-based by design	Score = $-\infty$	Anti-join
Interpretability	High (lift, confidence)	Low (latent space)	Medium (feature importance)
Feature richness	Association rules	Interaction history	21 features
Top signal	ECR (Lift $\times$ Conf $\times$ Price)	Latent dot product	w2v_cosine_sim
Training time	Seconds	5,804 s	167 s
Evaluation time	119.2 s	2.9 s	291.5 s
MAP@12	0.004517	0.006347	0.007466
Hit Rate@12	4.44%	4.83%	7.66%

TABLE X
HIERARCHICAL METRIC FRAMEWORK

Metric	Name	Type	Business Purpose	Statistical Test
Primary	Hit Rate@12	Binary	Measures algorithmic conversion propensity (did the user buy <i>any</i> recommended item?).	Pairwise Proportion Test (with Bonferroni Correction).
Secondary	IRR*	Continuous	Measures direct monetary value generated exclusively by algorithmic hits, proving true economic generation.	Welch's T-Test (equal_var=False).
Guardrail	SRM**	Categorical	Ensures the automated traffic router mathematically achieved the intended 34/33/33 split.	Chi-Square Test (scipy.stats.chisquare).

*Incremental Rec. Revenue — **Sample Ratio Mismatch

k : Total number of experimental arms (3 in this architecture)
 O_i : Observed count of unique users assigned to arm i
 E_i : Expected count based on target probability ($N \times p_i$)
 N : Total population size across all arms

A resulting P-value below the standard significance threshold ($\alpha = 0.05$) necessitates the immediate suspension of the experiment, as it proves the presence of SRM. Conversely, a P-value above 0.05 indicates insufficient evidence to reject the null hypothesis, thereby validating the integrity of the assignment router.

2) *CUPED Applicability Assessment*: Continuous metrics like Incremental Recommendation Revenue (IRR) suffer from high variance (σ^2), which diminishes Statistical Power ($1 - \beta$). To mitigate this in live environments, Controlled-experiment Using Pre-Experiment Data (CUPED) leverages historical covariates (e.g., pre-experiment monetary value from RFM clusters) to absorb explainable noise:

$$\hat{Y}_i^{CUPED} = Y_i - \theta(X_i - \bar{X})$$

Where:

Y_i : Observed continuous metric (IRR) for user i during the

test.

X_i : Pre-experiment covariate for user i (e.g., historical monetary average).

$\bar{X}$: Global mean of the pre-experiment covariate across all users.

θ : Optimal covariance multiplier calculated as $\left(\frac{\text{cov}(X, Y)}{\text{var}(X)} \right)$.

However, given our exceptionally large static holdout sample ($n = 216,479$), the Standard Error of the Mean (SEM) is natively suppressed. Therefore, mathematical CUPED transformation is bypassed in favor of raw economic evaluation via Welch's T-Test. We acknowledge CUPED as a mandatory architectural design for future live-environment variance reduction.

3) *Analysis of the Chi-Square Guardrail Output*: The test yielded $\chi^2 = 0.0744$ and $p = 0.9635$. Because $p \gg \alpha$ (0.05), the null hypothesis is retained. The router is mathematically validated and free of hashing bias, permitting causal statistical evaluation.

C. Formal Hypothesis Statement

1) *Primary Metric Hypotheses (Hit Rate Propensity)*: Evaluates if advanced architectures increase conversion probability compared to baseline rules.

$$\begin{cases} H_0 : p_{\text{treatment}} - p_{\text{control}} \leq 0 \\ H_1 : p_{\text{treatment}} - p_{\text{control}} > 0 \end{cases} \quad (3)$$

(Where p = proportion of converting users; $\alpha = 0.05$; Power $1 - \beta = 0.80$)

2) *Secondary Metric Hypotheses (Incremental Recommendation Revenue)*: Evaluates the continuous monetary accumulation derived exclusively from successful recommendations.

$$\begin{cases} H_0 : \mu_{\text{treatment}} - \mu_{\text{control}} \leq 0 \\ H_1 : \mu_{\text{treatment}} - \mu_{\text{control}} > 0 \end{cases} \quad (4)$$

(Where μ = Mean IRR per user in the respective arm)

D. Statistical Evaluation Methodology

1) Methodological Framework (Theory & Logic):

a) *Primary Metric: Pairwise Proportion Testing & Bonferroni Correction*: Because the Control Arm is concurrently tested against two independent variants (ALS and LightGBM), the Family-Wise Error Rate (FWER) inflates. To maintain strict mathematical rigor, the Bonferroni Correction is applied. We divide the global significance level ($\alpha = 0.05$) by $m = 2$ comparative pairs to secure the 95% confidence boundary.

Consequently, the adjusted significance threshold for the primary metric becomes $\alpha_{adj} = 0.025$. A treatment is only deemed statistically superior if its resulting P-value falls below this penalized boundary. The evaluation utilizes the two-sample Z-test for proportions, structured as:

$$Z = \frac{\hat{p}_t - \hat{p}_c}{\sqrt{\hat{p}_{pool}(1 - \hat{p}_{pool}) \left(\frac{1}{n_t} + \frac{1}{n_c} \right)}}$$

To satisfy C-level reporting requirements, the exact magnitude of the algorithmic uplift must be quantified using a Confidence Interval (CI) representing the delta (Δ) between the proportions:

$$CI_{\Delta} = (\hat{p}_t - \hat{p}_c) \pm Z_{1-\alpha_{adj}/2} \sqrt{\frac{\hat{p}_t(1 - \hat{p}_t)}{n_t} + \frac{\hat{p}_c(1 - \hat{p}_c)}{n_c}}$$

Note: A two-sided 95% Confidence Interval is reported alongside the one-tailed hypothesis test as standard practice for executive-facing reporting. The CI quantifies the plausible range of the uplift delta, while the one-tailed P-value evaluates whether that delta is directionally positive. Both are mathematically valid and mutually complementary.

b) Secondary Metric: Welch's T-Test and Effect Size:

The continuous secondary metric (IRR) violates the equal variance assumption (homoscedasticity) required by a tradi-

tional Student's T-Test. Fashion retail data is heavily right-skewed by high-frequency "Whale" consumers generating asymmetrical monetary density. To mathematically correct for this variance and differing sample sizes across arms, **Welch's T-Test** is mandated:

$$t = \frac{\bar{X}_t - \bar{X}_c}{\sqrt{\frac{s_t^2}{n_t} + \frac{s_c^2}{n_c}}}$$

While the resulting P-value confirms statistical significance, it does not convey the magnitude of the economic shift. In an A/B test exhibiting unequal variances, pooling the standard deviations creates a mathematically biased effect size. Therefore, **Glass's Delta** (Δ) is utilized instead of Cohen's d . Glass's Delta divides the mean difference strictly by the Control Group's standard deviation (s_c), anchoring the effect size to the stable baseline reference:

$$\Delta_{Glass} = \frac{\bar{X}_t - \bar{X}_c}{s_c}$$

2) Pre-Execution Artifact Analysis:

a) *Data Architecture & Schema Analysis*: Prior to executing the statistical engine, the physical structure of the experimental artifacts must be validated to ensure deterministic relational joins. The schema extraction yields the following architectural footprint across the evaluation environment in table XI:

b) *Architectural Normalization & Artifact Export*: Analysis reveals three critical dependencies required before statistical execution:

- 1) **ALS Type Misalignment**: `ALS_predictions` uses `Int64`. Requires downcast to `Int32` during ingestion to prevent memory bloat.
- 2) **Missing LightGBM Ranks**: `lgbm_test_features` lacks `Top-K` ordering.
- 3) **Dynamic MBA Overload**: Computing MBA rules concurrently during the statistical loop causes RAM failure.

⇒ **Standardization Protocol**: We append new code blocks to Phase 2 notebooks to export `MBA_predictions.parquet` and `lgbm_predictions.parquet`. All three predictive arms are standardized to an identical, compressed schema: `[customer_id (Int32), article_id (Int32), rank (Int8)]`.

3) Simulated RCT Execution & Statistical Inference:

a) *Statistical Engine Implementation*: To definitively evaluate the algorithmic uplift, the statistical engine will fuse the master assignment router, the unified predictive artifacts, and the historical ground truth.

Data Fusion & Evaluation Logic

- 1) **Isolate & Map**: The prediction artifacts (MBA, ALS,

TABLE XI
EXPERIMENTAL ARTIFACT SCHEMA

Artifact	Rows	Primary Keys	Target Value	Purpose / Status
ab_traffic_allocation	216,479	customer_id (Int32)	experimental_arm	Master router. (Ready)
test_processed	913,103	customer_id, article_id	price (Float64)	Ground truth. (Ready)
ALS_predictions	2.59M	customer_id, article_id	rank (Int64)	Trt A vectors. (Requires Int32 Downcast)
lgbm_test_features	17.0M	customer_id, article_id	label, features	Unscored pairs. (Deprecated)
articles_processed	105K	article_id (Int32)	index_name	Master catalog. (Ready)
MBA_predictions	-	customer_id, article_id	rank (Int8)	Control vectors. (Generate via Export)
lgbm_predictions	-	customer_id, article_id	rank (Int8)	Trt B vectors. (Generate via Export)

LightGBM) are strictly filtered to contain only the users assigned to their respective experimental arms (Control, Treatment A, Treatment B).

2) **Ground Truth Intersection:** The unified prediction matrix is joined against `test_processed.parquet` using `[customer_id, article_id]` as the composite key. A successful intersection represents an empirical “Hit”.

3) **User-Level Aggregation:**

- **Hit Rate Propensity (Binary):** A user is assigned a 1 if the intersection yields ≥ 1 item, otherwise 0.
- **Price-Weighted Hit Score (Continuous):** The price column from the ground truth is summed for all intersecting items per user. Non-converting users receive a strict 0.0.

Mathematical Formulation (Programmatic Implementation)

1) **Primary Metric (Proportion Z-Test):**

To adhere to the Bonferroni Correction ($\alpha_{adj} = 0.025$), the critical Z-value for the 95% Confidence Interval is calculated via the inverse Cumulative Distribution Function (CDF):

$$Z_{crit} = \Phi^{-1}(1 - \alpha_{adj}/2) \approx 2.2414$$

The Confidence Interval for the delta ($\Delta = \hat{p}_{treatment} - \hat{p}_{control}$) is computed using the unpooled standard error:

$$SE_{\Delta} = \sqrt{\frac{\hat{p}_t(1 - \hat{p}_t)}{n_t} + \frac{\hat{p}_c(1 - \hat{p}_c)}{n_c}}$$

$$CI = \Delta \pm (Z_{crit} \times SE_{\Delta})$$

2) **Secondary Metric (Welch’s T-Test & Cohen’s**

d): The continuous arrays are evaluated using `scipy.stats.ttest_ind(equal_var=False, alternative='greater')`. To measure standard deviation shifts independent of sample size, Cohen’s d is computed:

$$d = \frac{\bar{X}_t - \bar{X}_c}{\sqrt{\frac{s_t^2 + s_c^2}{2}}}$$

b) Empirical Outcome Analysis: The statistical engine successfully evaluated the simulated interactions of 216,479 unique test subjects against the historical ground truth. To ensure maximum analytical clarity, the empirical outputs are aggregated into three distinct evaluation matrices: separating the probability of conversion, the magnitude of economic generation, and the stratified behavioral variance across user clusters.

Statistical Aggregation Matrices

Table XII: Primary Metric - Hit Rate Propensity (Binary Conversion)

Evaluation Threshold: $\alpha_{adj} = 0.025$ (One-Tailed Z-Test)

Table XIII: Secondary Metric - Incremental Recommendation Revenue (IRR)

Evaluation Threshold: $\alpha_{adj} = 0.025$ (Welch’s T-Test)

Table XIV: Segment-Stratified Empirical Hit Rates

Analysis of variance across pre-defined RFM clusters.

Analytical Interpretations & Statistical Implications

I. Treatment A (ALS) Paradigm: Quantifying “ROI Trap”

The empirical data from Treatment A perfectly encapsulates the “ROI Trap” - a scenario where a statistical test yields mathematical significance but fails to deliver material economic utility. For the **Primary Metric**, the **Collaborative Filtering** architecture successfully utilized **Latent Factorization** to drive a **Relative Uplift** of 9.36% in conversion

TABLE XII
PRIMARY METRIC - HIT RATE PROPENSITY

Experimental Arm	Size (N)	Hit Rate	Abs. Δ	Rel. Uplift	Z-Score	P-Value	95% CI	Result
Control (MBA)	73,586	4.3881%	-	-	-	-	-	Baseline
Treatment A (ALS)	71,496	4.7989%	+0.4108%	+9.36%	3.7380	9.27×10^{-5}	[0.16%, 0.66%]	Significant
Treatment B (LGBM)	71,397	7.5073%	+3.1193%	+71.09%	25.1523	6.66×10^{-140}	[2.84%, 3.40%]	Significant

TABLE XIII
SECONDARY METRIC - INCREMENTAL RECOMMENDATION REVENUE (IRR)

Experimental Arm	Mean IRR	Std. Dev.	Abs. Shift	Welch's T	P-Value	Glass's Δ	Result
Control (MBA)	0.001360	0.007174	-	-	-	-	Baseline
Treatment A (ALS)	0.001439	0.007897	+0.000079	1.9874	2.34×10^{-2}	0.0110	Significant
Treatment B (LGBM)	0.002671	0.010267	+0.001312	28.1194	9.66×10^{-174}	0.1828	Significant

TABLE XIV
SEGMENT-STRATIFIED EMPIRICAL HIT RATES

Segment	Control	Trt A (ALS)	ALS Rel. Δ	Trt B (LGBM)	LGBM Rel. Δ
Mature Champions	4.3359%	5.0625%	+16.76%	7.9483%	+83.31%
Mature Newcomers	4.7848%	3.8701%	-19.11%	6.5616%	+37.13%
Young At-Risk	4.4572%	2.9616%	-33.55%	6.1375%	+37.69%
Young Core	4.2204%	4.6514%	+10.21%	6.4903%	+53.78%
Young Whales	4.4231%	4.9566%	+12.06%	7.7705%	+75.68%

propensity, achieving a definitive **Z-Score** of 3.7380.

Critically, following deduplication normalization, ALS also managed to clear the **Bonferroni-corrected threshold** ($\alpha_{adj} = 0.025$) for the **Secondary Metric**, returning a **Welch's T-Test P-Value** of 0.0234. However, despite achieving formal statistical significance across both axes, the economic magnitude is practically negligible. The **Absolute IRR Shift** (+0.000079) equates to a microscopic **Glass's Delta** ($\Delta = 0.0110$).

Unlike the LightGBM model which utilized **Expected Cross-Sell Revenue (ECR)** pricing thresholds and **Wallet-Matching**, the pure Matrix Factorization of ALS blindly optimized for structural co-occurrence. This made it fatally susceptible to **Popularity Bias**. It marginally increased the *frequency* of purchases (Hit Rate), but cannibalized margin density by recommending cheaper, high-velocity basics, proving that mathematical accuracy in a latent space does not natively guarantee increased basket scale.

II. Simpson's Paradox and Segment Harm

An examination of **Table 3** reveals a critical systemic flaw within the ALS architecture. The aggregate +9.36% Hit Rate uplift reported in Table 1 is mathematically misleading - it is an artifact of **Simpson's Paradox**.

While ALS generated positive lifts for high-frequency

segments like **Mature Champions** (+16.76%) and **Young Whales** (+12.06%), it actively destroyed value for dormant users. **Young At-Risk** users subjected to ALS converted at a rate **33.55% lower** than the MBA Control, and **Mature Newcomers** dropped by **19.11%**. Mechanically, Collaborative Filtering fails on lapsed/new users because their sparse interaction history is insufficient to anchor meaningful latent vectors. Consequently, ALS feeds them generic popularity lists that perform markedly worse than the demographically-aware **Association Rules** utilized by the Control arm.

III. Treatment B (LightGBM) Dominance: Universal Optimization

The performance of Treatment B establishes **Gradient Boosting** as the unequivocally superior algorithmic architecture. On the **Primary Metric**, LightGBM achieved an astronomical **Z-Score** of 25.1523, equating to a **P-value** approaching absolute zero (6.66×10^{-140}). The algorithm increased the conversion probability by a staggering **Relative Uplift** of 71.09%, with a tightly constrained **95% Confidence Interval** that operates entirely above a 2.8% absolute gain.

Crucially, LightGBM successfully cleared the dual-metric optimization hurdle while demonstrating universal segment superiority. The **Secondary Metric** demonstrates that the

Incremental Recommendation Revenue (IRR) effectively doubled (from 0.001360 to 0.002671). The **Welch’s T-Score** of 28.1194 mathematically confirms this systemic shift. The calculated **Glass’s Delta** ($\Delta = 0.1828$) represents a highly tangible, operationally impactful effect size, proving the revenue shift scales robustly relative to the control group’s baseline variance.

IV. Validation of the Anti-Leakage Methodology

These statistical milestones must be contextualized within the **Anti-Join Methodology** enforced during Phase 2. Because trivial repurchases were strictly barred from the candidate sets, the calculated **Uplifts** represent 100% pure **Novel Discovery**. The evaluation proves that advanced, multi-dimensional feature matrices possess the operational capacity to stimulate genuine, previously dormant consumer demand without relying on historical repurchasing crutches.

E. Business Impact & Deployment Decision

1) *ROI Trap Analysis & Annualized Revenue Projection*: The primary function of a Data-Driven Marketing strategy is to maximize the **Return on Investment (ROI)**. The “**ROI Trap**” occurs when an organization deploys a statistically significant algorithm whose computational infrastructure costs vastly exceed its marginal revenue generation. To make an objective deployment decision, the empirical **Absolute IRR Shift** calculated in Section 4 must be projected into an annualized economic forecast.

The simulation evaluated an active cohort of 216,479 users over a strictly bounded 28-day evaluation window. By utilizing the total segmented customer base (1,338,785 users) mapped during the RFM classification phase, the **Annualized Incremental Revenue** is forecasted utilizing the following scaling formula:

$$\text{Projected Annual Incr. Revenue} = (\Delta_{\mu} \times N_{Total}) \times \left(\frac{365}{28} \right) \quad (5)$$

Where elements are defined as:

- Δ_{μ} : **Absolute IRR Shift** (Continuous Incremental Recommendation Revenue per user).
- N_{Total} : **Total Addressable Market** (1.33 million active users).
- $365/28$: **Temporal Annualization Multiplier** (≈ 13.035).

Table XV: Annualized Economic Projection

Note: Monetary values are expressed in Normalized Revenue Units (NRU), corresponding to the normalized price scale [0, 0.591] used in the retail dataset. Conversion to absolute currency requires the original dataset normalization scalar. For executive reporting, the directional ratio (LightGBM generating $\sim 16.6\times$ the yield of ALS) remains economically valid regardless of unit scale.

Inference Cost Anchor:

To complete the ROI evaluation, the projected annual incremental yield of +22,897 NRU must be weighed against LightGBM’s inference costs. In production, this is measured in cloud compute units per batch scoring run (e.g., GPU/CPU-hour pricing per million candidate pairs). The scale of the yield differential - 22,897 NRU for LightGBM versus ALS’s negligible 1,379 NRU - provides a massive economic margin of safety. Deployment is strictly justified provided that monthly inference infrastructure costs remain below the annualized yield divided by 12 ($\approx 1,908$ NRU/month). This threshold must be validated against live cloud billing data prior to enterprise-wide scaling.

2) *Segment-Stratified Interpretation (RFM Lens)*: To understand *why* LightGBM overwhelmingly succeeded where ALS triggered the ROI Trap, the empirical metrics must be viewed through the RFM Cluster definitions established in Phase 1.

The ALS Simpson’s Paradox:

The aggregate Hit Rate uplift (+9.36%) produced by ALS masks severe structural failures. As demonstrated in the Segment Stratification matrix, ALS’s aggregate positive performance is entirely driven by high-frequency segments (Mature Champions: +16.76%, Young Whales: +12.06%). Shockingly, ALS actively harmed dormant segments: **Young At-Risk** users converted at a rate **33.55% lower** than the Control arm, and **Mature Newcomers** dropped by **19.11%**.

This phenomenon is caused by **Popularity Bias**. Because Collaborative Filtering treats all interactions uniformly within a latent matrix, it struggles with cold or lapsed users whose sparse history cannot anchor meaningful vectors. Consequently, ALS feeds them generic, cheap basics that fail to convert. It blindly optimizes for structural co-occurrence, entirely missing the margin constraints of the consumer.

The LightGBM Prescriptive Personalization:

Conversely, LightGBM generated positive uplifts in *every single segment*, effectively solving the sparse-history deficit. By incorporating **Wallet Matching** (user_avg_price) and **Expected Cross-Sell Revenue (ECR)** thresholds directly into the decision trees, the ranker actively optimized price acceptance:

- **The “Whale” Optimization**: For **Young Whales** (who represent 54.35% of total revenue), LightGBM recognized their historically high user_avg_price and algorithmically filtered out low-margin basics, driving an astonishing 75.68% Hit Rate uplift while pushing the mean IRR to 0.002836.
- **The “At-Risk” Reactivation**: For **Young At-Risk** consumers, LightGBM successfully avoided “sticker shock” by downgrading premium candidates, routing demographically appropriate items that secured a +37.69% Hit Rate reversal over the Control baseline.

TABLE XV
ANNUALIZED ECONOMIC PROJECTION

Architecture	Abs. IRR Shift (Δ_μ)	Stat. Validity	Proj. Annual Yield	Inference Cost
ALS (CF)	+0.000079	Passed ($p < \alpha_{adj}$)	+1,379 NRU	Medium (Latent Matrix Mult.)
LightGBM (L2R)	+0.001312	Passed ($p \ll \alpha_{adj}$)	+22,897 NRU	High (Feature Trees)

3) *C-Level Prescriptive Strategy & Deployment Mandate:*

Based on the statistical, economic, and segment-stratified evidence compiled throughout this Randomized Controlled Trial, the following strategic mandates are issued:

a) *Deprecate ALS & Transition to Supervised Learning-to-Rank:* The organization must immediately deprecate purely implicit matrix factorization (ALS) as a primary ranking engine. ALS successfully triggers interaction density but actively harms margin density and misfires on sparse-history users. The retail catalog exhibits a 90-day shelf life, mandating the use of Supervised Learning-to-Rank architectures capable of utilizing dynamic, multi-dimensional pricing and demographic features.

b) *Enforce “Novel Discovery” Metrics System-Wide:* The success of the LightGBM model was predicated on an **Anti-Join Architecture** that strictly blocked trivial repurchases during candidate generation. Marketing KPIs must reflect this paradigm shift. Replace Gross Hit Rate with **Net-New Hit Rate**, and substitute standard AOV with **Incremental Recommendation Revenue (IRR)**. These metrics measure true demand generation rather than artificially inflated repurchase tracking.

c) *Deploy Dual-Engine Prescriptive Logic:* The final production environment will utilize a bipartite algorithmic approach tailored to computational latency constraints:

- **Primary Navigation & Homepage (LightGBM):** Route traffic through the LightGBM L2R engine to govern the largest IRR opportunity. **Execution Protocol:** Initiate a **20% Phased Rollout**. Allocate 20% of live homepage traffic to LightGBM while monitoring real-time guardrail metrics (page latency, cart-to-checkout drop rate, and app crash rate) for a minimum of 14 days (capturing two full weekly seasonality cycles). Execute a 100% rollout only upon guardrail clearance.
- **Shopping Cart & Checkout (MBA Baseline):** Retain the MBA FP-Growth Baseline utilizing the Expected Cross-Sell Revenue (ECR) heuristic. Because LightGBM requires heavy batch feature engineering, it is ideal for asynchronous Homepage generation. Conversely, MBA Association Rules (N-to-1) execute via instant $O(1)$ dictionary lookups, making them the computationally optimal choice for real-time, impulse-buy generation at checkout.